📘 2. GOC (ADVANCED LEVEL 🔥)

1. Inductive vs Resonance (CORE CONCEPT)

Key Rule:

Resonance effect > Inductive effect

👉 Exception:

Distance badhne par inductive dominate kar sakta hai

2. Hyperconjugation (ADV THINKING)

Alpha hydrogen required

More alpha H → more stable

Example: $(CH_3)_3C^+$ most stable

3. Carbocation Stability (FULL LOGIC)

Order: 3° > 2° > 1° > CH_3^+

But actual rule: Resonance stabilized > hyperconjugation stabilized

👉 Benzyl, allyl > 3°

4. Carbanion Stability

Opposite trend: CH_3^- > 1° > 2° > 3°

But:

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Electron withdrawing group increases stability

5. Acid Strength (ADV LOGIC)

Depends on:

Stability of conjugate base

Resonance

Inductive effect

Hybridization

Order: sp > sp2 > sp3

👉 More s-character → more acidic

6. Reaction Intermediates Stability

Carbocation: Resonance > 3° > 2° > 1°

Free radical: Same trend but less pronounced

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